

AN UNVERIFIED AI-GENERATED CANDIDATE PROOF FOR THE NONSEPARABLE CASE OF KADISON'S TRACE-VECTOR BASIS PROBLEM FOR TYPE II_1 FACTORS

ABSTRACT. In 1967, Kadison asked whether every type II_1 factor has an orthonormal basis, with respect to its trace, consisting of unitaries. The case in which $L^2(M, \tau)$ is separable was recently settled affirmatively.

This manuscript presents an AI-generated and independently unverified candidate proof for the complementary nonseparable case in the tracial standard representation. More precisely, the candidate argument claims that if M is a type II_1 factor with faithful normal tracial state τ and

$$\text{dens } L^2(M, \tau) > \aleph_0,$$

then $L^2(M, \tau)$ admits a Hilbert orthonormal basis whose vectors are represented by self-adjoint unitaries in M , and hence an orthonormal basis consisting of trace vectors for the left action of M .

If correct, this argument, together with the previously established separable case, would give an affirmative solution to Kadison's 1967 problem in the tracial standard representation. The candidate proof is based on a relative norming lemma under small-density constraints, a finite-layer certified-background formalism designed to keep Hilbert-space projections represented by bounded operators in the ambient factor, and a transfinite extension over the density character of $L^2(M, \tau)$.

The mathematical argument has not been independently verified. This manuscript is released solely to invite careful scrutiny, correction, and possible further collaboration by specialists.

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1. OUTLINE OF THE PROOF

This section records the structure of the proof and fixes the main pieces of notation used below. The proof is organized around the real Hilbert space

$$H_0(P) = L^2(P, \tau)_{\text{sa}} \ominus \mathbb{R}1$$

associated with a finite von Neumann algebra P . By Proposition 3.1 and Lemma 3.2, it is enough to construct an orthonormal basis of $H_0(M)$ consisting of trace-zero symmetries in M . Adding the vector 1 then gives a complex Hilbert-space orthonormal basis of $L^2(M, \tau)$ represented by self-adjoint unitaries, and hence by trace vectors.

The main new ingredient in the nonseparable case is the relative norming lemma, Lemma 4.4. Suppose $N \subset M$ has smaller L^2 -density than M , and suppose that $r = r^* \in M$ is orthogonal to $L^2(N)_{\text{sa}}$ and to a prescribed set Γ of fewer than $\text{dens } L^2(M)$ self-adjoint constraints. The lemma constructs a trace-zero symmetry $s \in M$ which remains orthogonal to N and to Γ , and which satisfies the quantitative norming identity

$$\langle r, s \rangle_2 = \frac{\|r\|_2^2}{\|r\|_\infty}.$$

The proof uses a Krein–Milman extreme-point argument in a compact convex set of positive contractions. The only place where nonseparability enters is the small-constraint perturbation lemma, which uses the strict density gap $\text{dens } L^2(N) < \text{dens } L^2(M)$ to find a nonzero perturbation in a corner while annihilating all prescribed constraints.

The second technical point is that, for an uncountable orthonormal family, the Hilbert-space projection onto an arbitrary subfamily need not be represented by an element of M . Sections 5 and 6 introduce finite-layer certified backgrounds to avoid this problem. A completed block

$$L^2(Q)_{\text{sa}} = L^2(A)_{\text{sa}} \oplus \overline{\text{span}}_{\mathbb{R}} T$$

has the bounded projection formula

$$P_T(x) = E_Q(x) - E_A(E_Q(x)), \quad x = x^* \in M.$$

Finite-layer certificates are built by adjoining such blocks one layer at a time. The strong closure lemma, Proposition 6.4, says that every small subfamily of a certified background can be enlarged to a still small certified subbackground. This is the bookkeeping device which guarantees that all projections used later are bounded elements of M .

The countable extension lemma, Lemma 7.2, is the basic construction step. Starting from a small subalgebra N , a certified background Ω orthogonal to N , and countable sets of new elements to absorb, it constructs a separable completed block

$$L^2(R)_{\text{sa}} = L^2(C)_{\text{sa}} \oplus \overline{\text{span}}_{\mathbb{R}}(\Omega_0 \cup U),$$

where $C \subset N$, the set $\Omega_0 \subset \Omega$ is closed and certified, and U is a countable orthonormal family of trace-zero symmetries orthogonal to both N and Ω . The proof repeatedly processes all eventual self-adjoint $*$ -polynomial words. For a word y one writes

$$b(y) = y - E_N(y) - P_\Omega(y).$$

Whenever the residual part of $b(y)$ outside the already chosen span is nonzero, Lemma 4.4 supplies a new symmetry which reduces the squared distance by

$$\frac{\|r\|_2^4}{\|r\|_\infty^2}.$$

The bookkeeping is arranged so that each word is visited at times n_k with $\sum_k n_k^{-1} = \infty$; the elementary bound $\|r\|_\infty \leq K_y \sqrt{n_k}$ then forces the residual distance to converge to zero.

Proposition 8.3 upgrades the countable extension lemma to all cardinals $\lambda < \text{dens } L^2(M)$ by induction on λ . The proof uses no regularity assumption on the ambient density character; the needed cardinal estimates are recorded in Remark 8.2. At successor stages one certifies the old background together with the old newly constructed layer and applies the induction hypothesis again. At limit stages one takes unions of nested completed blocks and uses Lemma 8.1.

Finally, Proposition 9.1 converts the certified extension theorem into a relative basis extension property: if $N \subset M$ has smaller density and $H_0(N)$ already has a symmetry basis, then after adjoining any set of at most that density many self-adjoint elements one obtains a larger subalgebra P whose $H_0(P)$ again has a symmetry basis extending the old one. The main nonseparable theorem is then a transfinite recursion over a dense family $(a_\alpha)_{\alpha < \kappa}$ in $H_0(M)$, where $\kappa = \text{dens } L^2(M, \tau)$. Starting from a separable diffuse abelian subalgebra with its Walsh symmetry basis, the recursion builds nested subalgebras N_α and nested symmetry bases B_α , ensuring that $a_\alpha \in N_{\alpha+1}$. The union of the B_α 's is therefore an orthonormal basis of $H_0(M)$, and Lemma 3.2 completes the proof.

2. INTRODUCTION AND PROBLEM STATEMENT

Let M be a finite von Neumann algebra equipped with a faithful normal tracial state τ . Throughout the paper, all traces are normalized unless explicitly stated otherwise. We denote by $L^2(M, \tau)$ the Hilbert space completion of M for the norm $\|x\|_2 = \tau(x^*x)^{1/2}$. For $x \in M$, its image in $L^2(M, \tau)$ will be denoted by $\widehat{x}$. We write $M_{\text{sa}} = \{x \in M : x = x^*\}$ for the self-adjoint part of M . We also denote by $L^2(M, \tau)_{\text{sa}}$ the real Hilbert space obtained as the $\|\cdot\|_2$ -closure of M_{sa} in $L^2(M, \tau)$. We use the convention that the inner product is linear in the first variable:

$$\langle \widehat{x}, \widehat{y} \rangle_2 = \tau(y^*x), \quad x, y \in M.$$

When no confusion is possible, we identify an element $x \in M$ with its vector $\widehat{x} \in L^2(M, \tau)$. Throughout the paper, M acts on $L^2(M, \tau)$ by left multiplication. A vector $\xi \in L^2(M, \tau)$ is called a *trace vector* for this action if

$$\langle x\xi, \xi \rangle_2 = \tau(x), \quad x \in M.$$

By an *orthonormal basis* we always mean a Hilbert-space orthonormal basis, namely an orthonormal family whose closed linear span is the whole Hilbert space. In the nonseparable case such a basis may be uncountable. In this paper, “nonseparable” refers to the Hilbert-space density character $\text{dens } L^2(M, \tau)$, not to an arbitrary representation of M .

We also denote by $\mathcal{U}(M)$ the unitary group of M :

$$\mathcal{U}(M) = \{u \in M : u^*u = uu^* = 1\}.$$

A self-adjoint unitary will be called a *symmetry*.

Notation for families. If $N \subset M$ is a von Neumann subalgebra and $F \subset M$, the notation $E_N(F) = 0$ means that $E_N(f) = 0$ for every $f \in F$. If $F, G \subset L^2(M, \tau)$, or if one of them is a closed subspace, then $F \perp G$ means that every element of F is orthogonal to every element of G . We write $F \dot{\cup} G$ for a disjoint union and $\mathcal{P}(F)$ for the power set of F . If $\mathcal{P}_0 \subset M$, then $*\text{-alg}_{\mathbb{Q}+i\mathbb{Q}}(\mathcal{P}_0)$ denotes the unital $*$ -algebra over $\mathbb{Q} + i\mathbb{Q}$ generated by $\mathcal{P}_0$.

In 1967, at the Baton Rouge conference [Kad67], Kadison asked the trace-vector orthonormal basis problem for II_1 factors in the context of the natural Hilbert-space representations of group and group-measure-space factors¹. We formulate the question in the tracial standard representation:

¹Ge records Kadison's Problem 13 and states that the two preceding group/GMS realization questions have negative answers while the trace-vector basis question remains open; see [Ge03, p. 622, Problem 13]. See also [Pet20, Problem S.2].

Problem 2.1 (Kadison, tracial standard form). *Let M be a type II_1 factor with faithful normal tracial state τ . Does $L^2(M, \tau)$, for the left action of M , admit a Hilbert orthonormal basis consisting of trace vectors?*

In the tracial standard representation, Problem 2.1 is also naturally reformulated as a *unitary basis problem*. Indeed, if $u \in \mathcal{U}(M)$, then

$$\langle x\hat{u}, \hat{u} \rangle_2 = \tau(u^*xu) = \tau(x), \quad x \in M,$$

so $\hat{u}$ is a trace vector. Conversely, as recalled in Proposition 3.1, every trace vector in $L^2(M, \tau)$ is represented by a unitary in M . Thus, Problem 2.1 is equivalent to asking whether $L^2(M, \tau)$ has a Hilbert orthonormal basis consisting of unitaries in M .

The separable case of this problem was settled in [HTZ26]: using a finite-dimensional norming lemma derived from the noncommutative Lyapunov theorem of Akemann and Weaver [AW03], it was shown that every diffuse finite von Neumann algebra with separable L^2 -space admits a Hilbert orthonormal basis consisting of symmetries.

The purpose of the present paper is to prove the complementary nonseparable case. This case requires a different mechanism. One must extend bases through subalgebras of smaller density character while controlling orthogonality to previously chosen, possibly very large, backgrounds. The finite-layer certified background formalism records enough von Neumann algebraic structure to ensure that the relevant Hilbert-space projections remain bounded elements of M . This allows a transfinite extension argument over the density character of $L^2(M, \tau)$.

Organization. Section 3 contains the reduction from trace vectors to unitaries and from trace-zero symmetry bases to full complex bases. Sections 4–8 develop the relative extension machinery needed in the nonseparable setting. Section 9 proves the nonseparable theorem. Section 10 records the consequence obtained by combining the present theorem with the separable result of [HTZ26].

3. PRELIMINARIES AND REDUCTIONS

For a finite von Neumann algebra (P, τ) , we shall write

$$H_0(P) = L^2(P, \tau)_{\text{sa}} \ominus \mathbb{R}1 = \{\xi \in L^2(P, \tau)_{\text{sa}} : \langle \xi, 1 \rangle_2 = 0\}.$$

Thus, if $x = x^* \in P$, then $\hat{x} \in H_0(P)$ exactly when $\tau(x) = 0$. We also write

$$\mathcal{S}_0(P) = \{s \in P : s = s^* = s^{-1}, \tau(s) = 0\}$$

for the set of trace-zero symmetries in P . When P is a von Neumann subalgebra of a fixed finite von Neumann algebra, these notations are always understood with respect to the restricted trace.

We shall use the standard finite von Neumann algebra facts that the tracial standard representation has commutant given by right multiplication, that a trace-preserving conditional expectation exists for every unital inclusion of finite von Neumann algebras and is the L^2 -orthogonal projection, and that Kaplansky's density theorem gives L^2 -density on bounded sets in the presence of a finite trace; see, for example, [Tak72, Tak02, KR97a, KR97b].

We first record the standard reduction from trace vectors to unitaries.

Proposition 3.1. *Let (M, τ) be a finite von Neumann algebra in its tracial standard representation on $L^2(M, \tau)$. A vector $\xi \in L^2(M, \tau)$ is a trace vector for M if and only if*

$$\xi = \hat{u}$$

for some unitary $u \in \mathcal{U}(M)$.

Proof. If $u \in \mathcal{U}(M)$, then for every $x \in M$,

$$\langle x\widehat{u}, \widehat{u} \rangle_2 = \tau(u^*xu) = \tau(x),$$

so $\widehat{u}$ is a trace vector.

Conversely, suppose that ξ is a trace vector. Let $\Omega = \widehat{1}$. Define

$$V_0 : M\Omega \rightarrow L^2(M, \tau), \quad V_0(x\Omega) = x\xi.$$

For $x \in M$,

$$\|V_0(x\Omega)\|_2^2 = \|x\xi\|_2^2 = \langle x^*x\xi, \xi \rangle_2 = \tau(x^*x) = \|x\Omega\|_2^2.$$

Thus V_0 extends to an isometry V on $L^2(M, \tau)$. Moreover, for $a, x \in M$,

$$V(ax\Omega) = ax\xi = aV(x\Omega),$$

so $V \in M'$. Recall the standard identification of the commutant in the tracial representation [Tak02]: in the tracial standard representation of M on $L^2(M, \tau)$, the commutant M' is the right regular representation of M : for each $a \in M$,

$$R_a\widehat{x} = \widehat{xa}, \quad x \in M,$$

and the map $a \mapsto R_a$ is a $*$ -anti-isomorphism from M onto M' . In particular $M' \cong M^{\text{op}}$ is finite. Hence every isometry in M' is unitary. Indeed, let τ' be the faithful normal finite trace on M' transported from M^{op} . If $v \in M'$ and $v^*v = 1$, then

$$\tau'(1 - vv^*) = \tau'(1) - \tau'(vv^*) = \tau'(v^*v) - \tau'(vv^*) = 0.$$

Since $1 - vv^* \geq 0$ and τ' is faithful, we get $vv^* = 1$. Thus V is a unitary element of M' . Identifying M' with the right regular representation of M , we get $V\Omega = \widehat{u}$ for some $u \in \mathcal{U}(M)$. But $V\Omega = \xi$. Therefore $\xi = \widehat{u}$. $\square$

The next lemma is the basic trace-zero reduction used in the nonseparable proof.

Lemma 3.2 (Reduction to trace-zero symmetries). *Let M be a II_1 factor. If $B \subset \mathcal{S}_0(M)$ is a Hilbert orthonormal basis of the real Hilbert space $H_0(M)$, then $\{\widehat{1}\} \cup \{\widehat{s} : s \in B\}$ is a Hilbert orthonormal basis of the complex Hilbert space $L^2(M, \tau)$. Moreover every vector in $\{\widehat{1}\} \cup \{\widehat{s} : s \in B\}$ is a trace vector for the left action of M .*

Proof. For self-adjoint $a, b \in M$, the number $\tau(ba)$ is real, since

$$\tau(ba) = \tau(ab) = \tau((ba)^*).$$

Thus real orthogonality inside $L^2(M, \tau)_{\text{sa}}$ is the same as complex Hilbert-space orthogonality for these vectors. Since B is an orthonormal basis of $H_0(M)$ and $B \subset \mathcal{S}_0(M)$, the family $\{1\} \cup B$ is orthonormal in $L^2(M, \tau)$.

Let $\mathcal{L}$ be the closed complex linear span of $\{1\} \cup B$. Since

$$L^2(M, \tau)_{\text{sa}} = \mathbb{R}1 \oplus H_0(M) = \overline{\text{span}_{\mathbb{R}}(\{1\} \cup B)},$$

we have $L^2(M, \tau)_{\text{sa}} \subset \mathcal{L}$. Hence $M_{\text{sa}} \subset \mathcal{L}$, and therefore

$$M = M_{\text{sa}} + iM_{\text{sa}} \subset \mathcal{L}.$$

Since M is dense in $L^2(M, \tau)$, it follows that $\mathcal{L} = L^2(M, \tau)$. Thus $\{1\} \cup B$ is an orthonormal basis of the complex Hilbert space $L^2(M, \tau)$.

If $s = s^* = s^{-1} \in M$, then for every $x \in M$,

$$\langle xs, s \rangle_2 = \tau(sxs) = \tau(xss) = \tau(x).$$

Thus every self-adjoint unitary is a trace vector, and so is 1. $\square$

4. A RELATIVE NORMING LEMMA UNDER SMALL-DENSITY CONSTRAINTS

Throughout Sections 4–9, M denotes a II_1 factor with faithful normal tracial state τ and

$$\kappa = \text{dens } L^2(M, \tau) > \aleph_0.$$

All von Neumann subalgebras are understood to be unital, except that corners such as eMe are taken with their natural unit e . We use density character in the Hilbert norm: $\text{dens } L^2(M)$ is the least cardinality of an L^2 -dense subset.

We shall use standard facts on finite von Neumann algebras without further comment: Kaplansky's density theorem, the trace-preserving normal conditional expectation onto a von Neumann subalgebra of a finite von Neumann algebra, and projection comparison in finite factors; see, for example, [Tak72, KR97b, Tak02].

Lemma 4.1 (Elementary L^2 -closure and cardinal facts). *Let (P, τ) be a finite von Neumann algebra.*

- (i) *Let $A_0 \subset P$ be a unital C^* -subalgebra and let $A = A_0'' = W^*(A_0)$. If A_0 has norm density at most μ , then $L^2(A, \tau)$ has density at most $\max\{\mu, \aleph_0\}$.*
- (ii) *If $(P_i)_{i \in I}$ is a directed increasing family of von Neumann subalgebras of P and*

$$P_0 = W^*\left(\bigcup_{i \in I} P_i\right),$$

then

$$L^2(P_0, \tau) = \overline{\bigcup_{i \in I} L^2(P_i, \tau)}^{\|\cdot\|_2}.$$

- (iii) *If $Q \subset P$ is a von Neumann subalgebra and $x \in P$ satisfies $\widehat{x} \in L^2(Q, \tau)$, then $x \in Q$.*
- (iv) *If $Y \subset P_{\text{sa}}$ is L^2 -dense in $L^2(P, \tau)_{\text{sa}}$, then $W^*(Y) = P$.*

Proof. For (i), choose a norm-dense subset of A_0 of cardinal at most $\max\{\mu, \aleph_0\}$ and close it under rational complex linear combinations, adjoints, and products. This gives a norm-dense $*$ -subalgebra of A_0 of cardinal at most $\max\{\mu, \aleph_0\}$. By Kaplansky's density theorem, the unit ball of A_0 is strong- $*$ dense in the unit ball of A . On bounded subsets of a finite von Neumann algebra, strong convergence in the standard representation implies L^2 -convergence, because

$$\|x_\alpha - x\|_2 = \|(x_\alpha - x)\widehat{1}\|_2.$$

Hence the unit ball of A has an L^2 -dense subset of cardinal at most $\max\{\mu, \aleph_0\}$, and scaling gives the assertion for $L^2(A)$.

For (ii), put

$$B = \overline{\bigcup_i P_i}^{\|\cdot\|}.$$

Then B is a C^* -algebra and $B'' = P_0$. By Kaplansky's density theorem, the unit ball of B is strong- $*$ dense in the unit ball of P_0 . Since $\bigcup_i P_i$ is norm dense in B , it follows that $\bigcup_i P_i$ is L^2 -dense in P_0 on bounded sets, and hence

$$L^2(P_0, \tau) = \overline{\bigcup_i L^2(P_i, \tau)}^{\|\cdot\|_2}.$$

For (iii), let $E_Q : P \rightarrow Q$ be the trace-preserving conditional expectation. Since E_Q is the L^2 -orthogonal projection onto $L^2(Q)$, the assumption implies $\widehat{x} = \widehat{E_Q(x)}$. Faithfulness of τ gives $x = E_Q(x) \in Q$.

For (iv), put $Q = W^*(Y)$. Then $L^2(Q)_{\text{sa}}$ is a closed real subspace of $L^2(P)_{\text{sa}}$ containing Y , hence $L^2(Q)_{\text{sa}} = L^2(P)_{\text{sa}}$. Therefore $L^2(Q) = L^2(P)$, and (iii) gives $Q = P$. $\square$

Lemma 4.2 (Density of nonzero corners). *Let M be a II_1 factor with $\text{dens } L^2(M) = \kappa > \aleph_0$. If $0 \neq e \in M$ is a projection, then*

$$\text{dens } L^2(eMe) = \kappa,$$

where eMe is equipped with its finite trace inherited from τ .

Proof. Clearly $\text{dens } L^2(eMe) \leq \kappa$. Let $\mu = \text{dens } L^2(eMe)$. Choose $n \in \mathbb{N}$ with $1/n \leq \tau(e)$. In a II_1 factor there are orthogonal projections $e_1, \dots, e_n$ with $\sum_i e_i = 1$ and $\tau(e_i) = 1/n$. There are also projections $f_i \leq e$ with $\tau(f_i) = 1/n$; these f_i need not be mutually orthogonal. Let v_i be partial isometries with $v_i^* v_i = f_i$ and $v_i v_i^* = e_i$. For $x \in M$,

$$e_i x e_j = v_i (v_i^* x v_j) v_j^*, \quad v_i^* x v_j \in f_i M f_j \subset eMe.$$

Thus M is contained in the finite linear span of the subspaces $v_i(eMe)v_j^*$. Each of these subspaces has L^2 -density at most μ , so

$$\kappa = \text{dens } L^2(M) \leq \max\{\mu, \aleph_0\}.$$

Since $\kappa > \aleph_0$, this inequality forces $\mu > \aleph_0$ and hence $\max\{\mu, \aleph_0\} = \mu$. Therefore $\kappa \leq \mu$, and the reverse inequality already observed gives $\mu = \kappa$. $\square$

Lemma 4.3 (Small-constraint perturbation). *Let $\Gamma \subset M_{\text{sa}}$ satisfy $|\Gamma| < \kappa$. If $0 \neq e \in M$ is a projection, then there exists a nonzero $h = h^* \in eMe$ such that*

$$\tau(\gamma h) = 0, \quad \gamma \in \Gamma.$$

Proof. Let $D = W^*(\{e\} \cup \Gamma)$. If $\lambda = |\Gamma|$, then the unital C^* -algebra generated by $\{e\} \cup \Gamma$ has norm density at most $\max\{\lambda, \aleph_0\} < \kappa$, because rational-coefficient $*$ -polynomials in the generators are norm dense in it. By Lemma 4.1(i),

$$\text{dens } L^2(D) < \kappa.$$

Hence also $\text{dens } L^2(eDe) < \kappa$.

By Lemma 4.2,

$$\text{dens } L^2(eMe) = \kappa.$$

Since

$$L^2(eMe) = L^2(eMe)_{\text{sa}} + iL^2(eMe)_{\text{sa}}$$

and similarly for eDe , the real Hilbert space $L^2(eMe)_{\text{sa}}$ has density κ , while $L^2(eDe)_{\text{sa}}$ has density $< \kappa$. Thus $L^2(eDe)_{\text{sa}}$ is a proper closed real subspace of $L^2(eMe)_{\text{sa}}$. Choose

$$\eta = \eta^* \in L^2(eMe)_{\text{sa}} \setminus L^2(eDe)_{\text{sa}}.$$

Since $(eMe)_{\text{sa}}$ is L^2 -dense in $L^2(eMe)_{\text{sa}}$ and $L^2(eDe)_{\text{sa}}$ is closed, we may choose $a = a^* \in eMe$ so close to η in L^2 -norm that

$$\hat{a} \notin L^2(eDe)_{\text{sa}}.$$

In particular,

$$a \in eMe \setminus eDe.$$

Let

$$h = a - E_D(a),$$

where $E_D : M \rightarrow D$ is the trace-preserving conditional expectation. Since $e \in D$ and $a = eae$, bimodularity of E_D gives

$$E_D(a) = E_D(eae) = eE_D(a)e \in eDe.$$

Thus $h \in eMe$. Moreover $h \neq 0$, because otherwise $a = E_D(a) \in eDe$.

Finally, for $\gamma \in \Gamma \subset D$, trace preservation and D -bimodularity of E_D give

$$\tau(\gamma a) = \tau(E_D(\gamma a)) = \tau(\gamma E_D(a)).$$

Therefore $\tau(\gamma h) = 0$ for every $\gamma \in \Gamma$. □

Lemma 4.4 (Large relative norming lemma). *Let $\kappa = \text{dens } L^2(M) > \aleph_0$. Let $N \subset M$ be a von Neumann subalgebra with $\text{dens } L^2(N) < \kappa$, and let $\Gamma \subset M_{\text{sa}}$ have cardinality $< \kappa$. Suppose $0 \neq r = r^* \in M$ satisfies*

$$E_N(r) = 0, \quad \tau(\gamma r) = 0 \quad (\gamma \in \Gamma).$$

Then there exists $s \in \mathcal{S}_0(M)$ such that

$$E_N(s) = 0, \quad \tau(\gamma s) = 0 \quad (\gamma \in \Gamma),$$

and

$$\langle r, s \rangle_2 = \frac{\|r\|_2^2}{\|r\|_\infty}.$$

Proof. Set $c = \|r\|_\infty$. Since $r \neq 0$, we have $c > 0$. Put

$$q = \frac{1 + r/c}{2}.$$

Then $0 \leq q \leq 1$. Choose a subset $\mathcal{D} \subset N_{\text{sa}}$ consisting of bounded operators, L^2 -dense in $L^2(N)_{\text{sa}}$, and satisfying

$$|\mathcal{D}| \leq \max\{\text{dens } L^2(N), \aleph_0\} < \kappa.$$

Consider the set

$$K = \left\{ 0 \leq x = x^* \leq 1 : \begin{array}{ll} \tau(yx) = \tau(yq) & (y \in \mathcal{D}), \\ \tau(\gamma x) = \tau(\gamma q) & (\gamma \in \Gamma), \\ \tau(rx) = \tau(rq) \end{array} \right\}.$$

It is nonempty because $q \in K$. It is ultraweakly compact and convex: indeed, it is the intersection of the ultraweakly compact positive unit ball of M with ultraweakly closed affine hyperplanes defined by normal functionals. By the Krein–Milman theorem [Con90], K has an extreme point. Let p be one.

We claim that p is a projection. Suppose not. Since $0 \leq p \leq 1$ and p is not a projection, the spectral projection of p corresponding to the open interval $(0, 1)$ is nonzero. Since

$$(0, 1) = \bigcup_{n=2}^{\infty} [1/n, 1 - 1/n],$$

there is some $0 < \varepsilon < 1/2$ such that

$$e = 1_{[\varepsilon, 1-\varepsilon]}(p)$$

is nonzero.

Apply Lemma 4.3 to the set

$$\mathcal{D} \cup \Gamma \cup \{r\}$$

and to the projection e . This set has cardinal $< \kappa$. We get a nonzero $h = h^* \in eMe$ such that

$$\tau(yh) = 0 \quad (y \in \mathcal{D}), \quad \tau(\gamma h) = 0 \quad (\gamma \in \Gamma), \quad \tau(rh) = 0.$$

After replacing h by a sufficiently small nonzero real scalar multiple, we may assume $\|h\|_\infty < \varepsilon$. Since e is a spectral projection of p , it commutes with p . Also $h = ehe$. Hence, with respect to the decomposition $eL^2(M) \oplus (1-e)L^2(M)$, the operators $p \pm h$ are block diagonal:

$$p \pm h = (pe \pm h) \oplus p(1-e).$$

On $eL^2(M)$ we have

$$\varepsilon e \leq pe \leq (1 - \varepsilon)e.$$

Since $h = h^* \in eMe$, we have

$$-\|h\|_\infty e \leq h \leq \|h\|_\infty e.$$

As $\|h\|_\infty < \varepsilon$, it follows that

$$0 \leq pe \pm h \leq e.$$

On $(1 - e)L^2(M)$, the operator $p(1 - e)$ is a positive contraction. Therefore

$$0 \leq p \pm h \leq 1.$$

The displayed orthogonality relations show that $p + h$ and $p - h$ satisfy all defining affine constraints of K . Thus $p \pm h \in K$, and

$$p = \frac{1}{2}(p + h) + \frac{1}{2}(p - h)$$

is a nontrivial convex decomposition in K , contradicting the extremality of p . Hence p is a projection.

Put $s = 2p - 1$. For $y \in \mathcal{D}$,

$$\tau(yq) = \frac{1}{2}\tau(y) + \frac{1}{2c}\tau(yr) = \frac{1}{2}\tau(y),$$

since $E_N(r) = 0$. Thus $\tau(ys) = 0$ for every $y \in \mathcal{D}$. If $z \in N_{\text{sa}}$, choose $y_i \in \mathcal{D}$ with $\|y_i - z\|_2 \rightarrow 0$. Since $s \in M$ is bounded, the functional $w \mapsto \tau(ws)$ is L^2 -continuous on $L^2(N)_{\text{sa}}$. Hence $\tau(zs) = 0$ for all $z \in N_{\text{sa}}$. Since $E_N(s) \in N_{\text{sa}}$, we may take $z = E_N(s)$. This gives

$$0 = \tau(E_N(s)s) = \tau(E_N(E_N(s)s)) = \tau(E_N(s)^2) = \|E_N(s)\|_2^2,$$

where we used that E_N is trace preserving and N -bimodular. Hence $E_N(s) = 0$. Similarly, for $\gamma \in \Gamma$,

$$\tau(\gamma q) = \frac{1}{2}\tau(\gamma) + \frac{1}{2c}\tau(\gamma r) = \frac{1}{2}\tau(\gamma),$$

so $\tau(\gamma s) = 0$. Since $1 \in N$, the identity $E_N(s) = 0$ also gives $\tau(s) = 0$. Therefore $s \in \mathcal{S}_0(M)$.

Finally, using the convention that the inner product is linear in the first variable and using traciality,

$$\begin{aligned} \langle r, s \rangle_2 &= \tau(sr) = \tau(rs) \\ &= \tau(r(2p - 1)) \\ &= 2\tau(rp) - \tau(r) \\ &= 2\tau(rq) - \tau(r) \\ &= 2 \left(\frac{1}{2}\tau(r) + \frac{1}{2c}\tau(r^2) \right) - \tau(r) = \frac{\|r\|_2^2}{\|r\|_\infty}. \end{aligned}$$

□

5. COMPLETED BLOCKS AND CERTIFIED BACKGROUNDS

Definition 5.1 (Completed block). A *completed block* in M is a triple

$$\mathfrak{B} = (A, Q, T),$$

where $A \subset Q \subset M$ are von Neumann subalgebras and $T \subset \mathcal{S}_0(M)$ is an orthonormal family such that

$$L^2(Q)_{\text{sa}} = L^2(A)_{\text{sa}} \oplus \overline{\text{span}_{\mathbb{R}} T}.$$

The sum is an orthogonal direct sum of real Hilbert spaces.

If a von Neumann subalgebra $N \subset M$ is specified, we say that $\mathfrak{B}$ is a *completed block over* N if, in addition,

$$A \subset N \quad \text{and} \quad E_N(t) = 0 \quad (t \in T).$$

In that case $T \perp L^2(N)_{\text{sa}}$, and in particular $T \perp L^2(A)_{\text{sa}}$.

In either case, the displayed decomposition implies $T \subset Q$: indeed $t \in T$ is an element of M whose L^2 -vector belongs to $L^2(Q)$, so Lemma 4.1(iii) gives $t \in Q$.

Lemma 5.2 (Projection formula). *If $\mathfrak{B} = (A, Q, T)$ is a completed block, then for every $x = x^* \in M$ the orthogonal projection of x onto $\overline{\text{span}_{\mathbb{R}} T}$ is*

$$P_T^{\mathfrak{B}}(x) = E_Q(x) - E_A(E_Q(x)).$$

In particular $P_T^{\mathfrak{B}}(x) \in M_{\text{sa}}$.

Proof. The map E_Q is the L^2 -orthogonal projection onto $L^2(Q)$ and E_A is the L^2 -orthogonal projection from $L^2(Q)$ onto $L^2(A)$. The completed-block decomposition identifies $L^2(Q)_{\text{sa}} \ominus L^2(A)_{\text{sa}}$ with $\overline{\text{span}_{\mathbb{R}} T}$. $\square$

We use finite certificates to record when projections onto large orthonormal families are represented by bounded operators. The word “finite” refers to the depth of the certificate tree; the individual layers may be uncountable.

Definition 5.3 (Finite-layer certificates and certified backgrounds). A *finite-layer certificate* $\mathfrak{c}$ is defined recursively, together with its underlying orthonormal family

$$\Omega(\mathfrak{c}) \subset \mathcal{S}_0(M)$$

and, for every $x = x^* \in M$, a bounded self-adjoint element $P_{\mathfrak{c}}(x) \in M_{\text{sa}}$.

The empty certificate. There is an empty certificate $\emptyset$ with

$$\Omega(\emptyset) = \emptyset, \quad P_{\emptyset}(x) = 0.$$

Adding one layer. Suppose that $\mathfrak{c}^-$ is a finite-layer certificate with

$$\Omega^- = \Omega(\mathfrak{c}^-).$$

Suppose also that $\mathfrak{c}_{\Theta}$ is a finite-layer certificate with

$$\Theta = \Omega(\mathfrak{c}_{\Theta}) \subset \Omega^-.$$

Let $U \subset \mathcal{S}_0(M)$ be an orthonormal family which is orthogonal to Ω^- . Assume that there are von Neumann subalgebras

$$C \subset R \subset M$$

such that

$$L^2(R)_{\text{sa}} = L^2(C)_{\text{sa}} \oplus \overline{\text{span}_{\mathbb{R}}(\Theta \cup U)}.$$

Then the finite data

$$\mathfrak{c} = (\mathfrak{c}^-, \mathfrak{c}_{\Theta}, C, R, U)$$

is a finite-layer certificate with underlying family

$$\Omega(\mathfrak{c}) = \Omega^- \dot{\cup} U.$$

We say that U is the *new layer* of $\mathfrak{c}$, and that it is a layer over the certified subbackground $(\Theta, \mathfrak{c}_{\Theta})$.

For $x = x^* \in M$, define

$$P_U^{\mathfrak{c}}(x) = (E_R(x) - E_C(E_R(x))) - P_{\mathfrak{c}_{\Theta}}(x),$$

and

$$P_{\mathfrak{c}}(x) = P_{\mathfrak{c}^-}(x) + P_U^{\mathfrak{c}}(x).$$

A *finite-layer certified background* is a pair $(\Omega, \mathfrak{c})$, where $\mathfrak{c}$ is a finite-layer certificate and $\Omega = \Omega(\mathfrak{c})$. When the certificate is fixed, we suppress it from the notation and write simply $P_{\Omega}(x)$ for $P_{\mathfrak{c}}(x)$.

A subfamily $\Theta \subset \Omega$ is called a *certified subbackground* when it is equipped with some finite-layer certificate $\mathbf{c}_\Theta$ satisfying $\Omega(\mathbf{c}_\Theta) = \Theta$. This certificate need not be a literal subcertificate of the fixed certificate for Ω ; it is part of the finite data being used. The adjective *finite-layer* refers only to the finite depth of the certificate. The individual layers may have arbitrary cardinality.

Remark 5.4 (Complexity of certificates). We define the complexity of a finite-layer certificate recursively by

$$\ell(\emptyset) = 0,$$

and, for

$$\mathbf{c} = (\mathbf{c}^-, \mathbf{c}_\Theta, C, R, U),$$

by

$$\ell(\mathbf{c}) = 1 + \ell(\mathbf{c}^-) + \ell(\mathbf{c}_\Theta).$$

All inductions on finite-layer certificates below are taken with respect to this finite complexity.

Lemma 5.5 (Boundedness of certified projections). *Let Ω be a finite-layer certified background. Then for every $x = x^* \in M$, the Hilbert-space projection $P_\Omega x$ onto $\overline{\text{span}}_{\mathbb{R}} \Omega$ belongs to M_{sa} . More precisely, $P_\Omega x$ is a finite sum of expressions obtained from x by trace-preserving conditional expectations appearing in the certificate.*

Proof. We prove the assertion by induction on the complexity of the finite certificate. The empty certificate is trivial.

Let

$$\mathbf{c} = (\mathbf{c}^-, \mathbf{c}_\Theta, C, R, U)$$

be a certificate obtained by adding one layer. Put

$$\Omega^- = \Omega(\mathbf{c}^-), \quad \Theta = \Omega(\mathbf{c}_\Theta), \quad \Omega = \Omega^- \dot{\cup} U.$$

By the induction hypothesis, $P_{\mathbf{c}^-}(x)$ is the Hilbert-space projection of x onto $\overline{\text{span}}_{\mathbb{R}} \Omega^-$, and $P_{\mathbf{c}_\Theta}(x)$ is the Hilbert-space projection of x onto $\overline{\text{span}}_{\mathbb{R}} \Theta$. Both belong to M_{sa} .

The completed-block decomposition

$$L^2(R)_{\text{sa}} = L^2(C)_{\text{sa}} \oplus \overline{\text{span}}_{\mathbb{R}}(\Theta \cup U)$$

and Lemma 5.2 show that

$$E_R(x) - E_C(E_R(x))$$

is the Hilbert-space projection of x onto $\overline{\text{span}}_{\mathbb{R}}(\Theta \cup U)$. Since $U \perp \Omega^-$ and $\Theta \subset \Omega^-$, this projection splits orthogonally as the sum of the projection onto $\overline{\text{span}}_{\mathbb{R}} \Theta$ and the projection onto $\overline{\text{span}}_{\mathbb{R}} U$. Hence

$$P_U^{\mathbf{c}}(x) = (E_R(x) - E_C(E_R(x))) - P_{\mathbf{c}_\Theta}(x)$$

is exactly the Hilbert-space projection of x onto $\overline{\text{span}}_{\mathbb{R}} U$. Therefore

$$P_{\mathbf{c}}(x) = P_{\mathbf{c}^-}(x) + P_U^{\mathbf{c}}(x)$$

is the Hilbert-space projection of x onto $\overline{\text{span}}_{\mathbb{R}} \Omega$. The displayed formula is a finite combination of trace-preserving conditional expectations and previously constructed projection formulas, so it belongs to M_{sa} . $\square$

Consequently, if two finite-layer certificates have the same underlying orthonormal family Θ , then their formulas give the same element of M_{sa} for every $x = x^* \in M$, since both compute the Hilbert-space projection of x onto $\overline{\text{span}}_{\mathbb{R}} \Theta$.

6. THE STRONG CLOSURE LEMMA FOR CERTIFIED BACKGROUNDS

The main technical issue is that an arbitrary subfamily of a certified background need not have bounded projection. The next proposition gives a canonical way to enlarge any small subfamily to a small certified subbackground.

Purpose of the closure construction. The closure operation below is used only to ensure the following point: whenever a small subfamily of a certified background is selected, it can be enlarged to a still small subfamily on which all relevant projection supports are closed. Consequently the Hilbert-space projections onto these enlarged subfamilies are represented by bounded elements of the ambient von Neumann algebra M . No additional model-theoretic property of M is being assumed.

For an orthonormal family T and $\xi \in L^2(M)$, write

$$\text{supp}_T(\xi) = \{t \in T : \langle \xi, t \rangle_2 \neq 0\}.$$

Lemma 6.1 (Countability of supports). *Let T be an orthonormal family in a Hilbert space H . For every $\xi \in H$,*

$$\text{supp}_T(\xi) = \{t \in T : \langle \xi, t \rangle \neq 0\}$$

is countable.

Proof. For $n \geq 1$, the set

$$F_n = \{t \in T : |\langle \xi, t \rangle| \geq 1/n\}$$

is finite by Bessel's inequality. Since

$$\text{supp}_T(\xi) = \bigcup_{n=1}^{\infty} F_n,$$

the support is countable. □

Here and below, for a regular cardinal χ , $H(\chi)$ denotes the set of all sets whose transitive closure has cardinality $< \chi$. We use only the standard elementary consequences of closing $H(\chi)$ under countably many finitary Skolem functions.

Lemma 6.2 (Skolem-hull closure fact). *Let χ be regular and let $\mathfrak{A}$ be a countable first-order expansion of $(H(\chi), \in)$ by finitary function symbols, including Skolem functions, constants for ω , and constants for all natural numbers. For $S \subset H(\chi)$, let $\text{Hull}_{\mathfrak{A}}(S)$ denote the Skolem hull generated by S together with the named constants. Then:*

- (i) $|\text{Hull}_{\mathfrak{A}}(S)| \leq |S| + \aleph_0$;
- (ii) $S_1 \subset S_2$ implies $\text{Hull}_{\mathfrak{A}}(S_1) \subset \text{Hull}_{\mathfrak{A}}(S_2)$;
- (iii) for every increasing chain $(S_i)_{i \in I}$,

$$\text{Hull}_{\mathfrak{A}}\left(\bigcup_i S_i\right) = \bigcup_i \text{Hull}_{\mathfrak{A}}(S_i);$$

- (iv) if $c \in \text{Hull}_{\mathfrak{A}}(S)$ is countable in the universe, then $c \subset \text{Hull}_{\mathfrak{A}}(S)$.

Proof. The first three assertions follow from the usual construction of the closure under countably many finitary Skolem functions; in the chain-continuity statement, every finite tuple from an increasing union is already contained in one member of the chain. For (iv), the hull is an elementary submodel of the Skolemized structure and contains every natural number. If $c = \emptyset$, there is nothing to prove. Assume that $c \neq \emptyset$. Since c is countable, $H(\chi)$ satisfies that there is a surjection $f : \omega \rightarrow c$. By elementarity and the Skolem functions, such an f belongs to the hull. For every $n \in \omega$, the value $f(n)$ is definable from f and n , and hence belongs to the hull. Thus $c \subset \text{Hull}_{\mathfrak{A}}(S)$. This is the standard Skolem-hull argument; see, for example, [Jech03, CK90]. □

Lemma 6.3 (Closed subblocks of a completed block). *Let $\mathfrak{B} = (A, Q, T)$ be a completed block. There is a closure operation $\text{cl}_{\mathfrak{B}}$ on subsets of T with the following properties:*

- (i) $S \subset \text{cl}_{\mathfrak{B}}(S)$;
- (ii) $\text{cl}_{\mathfrak{B}}(\text{cl}_{\mathfrak{B}}(S)) = \text{cl}_{\mathfrak{B}}(S)$;
- (iii) $|\text{cl}_{\mathfrak{B}}(S)| \leq |S| + \aleph_0$;
- (iv) $S_1 \subset S_2$ implies $\text{cl}_{\mathfrak{B}}(S_1) \subset \text{cl}_{\mathfrak{B}}(S_2)$;
- (v) for every increasing chain $(S_i)_{i \in I}$,

$$\text{cl}_{\mathfrak{B}}\left(\bigcup_i S_i\right) = \bigcup_i \text{cl}_{\mathfrak{B}}(S_i);$$

- (vi) if $S = \text{cl}_{\mathfrak{B}}(S)$, then there are von Neumann algebras $A_S \subset A$ and $Q_S \subset Q$ such that

$$(A_S, Q_S, S)$$

is a completed block.

Proof. We spell out the Skolem-hull construction. Choose a regular cardinal χ such that

$$M, A, Q, T, \tau, E_A, E_Q, \omega, [T]^{\leq \aleph_0}$$

belong to $H(\chi)$, and such that the graphs of the algebraic operations on M also belong to $H(\chi)$. We take χ large enough for all these requirements throughout the argument below.

Work in a countable first-order expansion $\mathfrak{A}_{\mathfrak{B}}$ of $(H(\chi), \in)$. The objects M, A, Q, T , the algebraic operations on M , and the conditional expectations E_A, E_Q are named as single predicates, function symbols, or set-valued parameters. The algebraic operations include the constants 0_M and 1_M , addition, multiplication, adjoint, and scalar multiplication by each $\lambda \in \mathbb{Q} + i\mathbb{Q}$. We do not add constants for all individual elements of M, A, Q , or T .

We include a constant for ω and constants for every natural number. Hence every Skolem hull $\mathcal{E}$ considered below satisfies

$$\omega \in \mathcal{E} \quad \text{and} \quad \omega \subset \mathcal{E}.$$

Every partial map used below is made into a total function on $H(\chi)$ by assigning an arbitrary fixed value outside its intended domain.

As a function symbol in the expansion, include the total support map

$$\sigma_{\mathfrak{B}}(x) = \text{supp}_T(E_Q(x) - E_A(E_Q(x))), \quad x = x^* \in M,$$

again with an arbitrary fixed value outside M_{sa} . By Lemma 6.1, for $x = x^* \in M$ the value $\sigma_{\mathfrak{B}}(x)$ is a countable subset of T , and hence belongs to $H(\chi)$.

The language is countable, since only countably many algebraic operations, conditional expectations, support maps, and named structural parameters have been added. Add countably many finitary Skolem functions for this countable structure. For $S \subset T$, let $\text{Hull}_{\mathfrak{B}}(S)$ be the Skolem hull generated by S , together with the fixed structural parameters just described, the constant ω , and the constants for all natural numbers. Define

$$\text{cl}_{\mathfrak{B}}(S) = T \cap \text{Hull}_{\mathfrak{B}}(S).$$

Thus the hull is generated from S , the natural numbers, and the fixed structural objects, but not from all individual elements of M, A, Q , or T .

By Lemma 6.2, the size, monotonicity, and chain-continuity assertions follow from the countability of the language and the finitariness of the Skolem functions. Idempotence follows because

$$\text{Hull}_{\mathfrak{B}}(S) \subset \text{Hull}_{\mathfrak{B}}(\text{cl}_{\mathfrak{B}}(S)) \subset \text{Hull}_{\mathfrak{B}}(\text{Hull}_{\mathfrak{B}}(S)) = \text{Hull}_{\mathfrak{B}}(S).$$

Assume now that $S = \text{cl}_{\mathfrak{B}}(S)$, and set $\mathcal{E} = \text{Hull}_{\mathfrak{B}}(S)$. Then $S = T \cap \mathcal{E}$. We shall use Lemma 6.2(iv): if $c \in \mathcal{E}$ is countable, then $c \subset \mathcal{E}$.

Define

$$A_S = W^*(A \cap \mathcal{E}), \quad Q_S = W^*(A_S, S).$$

We show that (A_S, Q_S, S) is a completed block. Let $w = w(a_1, \dots, a_m, s_1, \dots, s_n)$ be a bounded self-adjoint $*$ -word with coefficients in $\mathbb{Q} + i\mathbb{Q}$, where $a_i \in A \cap \mathcal{E}$ and $s_j \in S$. Since $S \subset T \subset Q$ and $A \subset Q$, we have $w \in Q$. Moreover, because the algebraic operations are included in the Skolemized structure, $w \in \mathcal{E}$. In the original block,

$$w = E_A(w) + P_T^{\mathfrak{B}}(w).$$

Since E_A is part of the structure, $E_A(w) \in A \cap \mathcal{E}$. Moreover

$$\text{supp}_T(P_T^{\mathfrak{B}}(w)) = \sigma_{\mathfrak{B}}(w) \in \mathcal{E}.$$

This set is countable. Since $\mathcal{E}$ contains ω and all natural numbers, Lemma 6.2(iv) gives

$$\sigma_{\mathfrak{B}}(w) \subset \mathcal{E}.$$

Hence

$$\sigma_{\mathfrak{B}}(w) \subset T \cap \mathcal{E} = S,$$

and therefore

$$P_T^{\mathfrak{B}}(w) \in \overline{\text{span}_{\mathbb{R}} S}.$$

Let $\mathcal{A}_S$ be the unital rational-coefficient algebraic $*$ -algebra generated by $A \cap \mathcal{E}$ and S , and let

$$B_S = \overline{\mathcal{A}_S}^{\|\cdot\|}$$

be its norm closure. Then

$$Q_S = W^*(A_S, S) = W^*(A \cap \mathcal{E}, S) = B_S''.$$

Kaplansky's density theorem applies to the C^* -algebra B_S : the self-adjoint part of the unit ball of B_S is strong- $*$ dense in the self-adjoint part of the unit ball of Q_S . Since $\mathcal{A}_S$ is norm dense in B_S , and since bounded strong convergence implies L^2 -convergence in a finite von Neumann algebra, the bounded self-adjoint rational-coefficient $*$ -words in $A \cap \mathcal{E}$ and S are L^2 -dense in $L^2(Q_S)_{\text{sa}}$. Hence

$$L^2(Q_S)_{\text{sa}} \subset L^2(A_S)_{\text{sa}} + \overline{\text{span}_{\mathbb{R}} S}.$$

The reverse inclusion is clear from the definition of Q_S , and the sum is orthogonal because $S \perp L^2(A)_{\text{sa}}$ and $A_S \subset A$. Thus

$$L^2(Q_S)_{\text{sa}} = L^2(A_S)_{\text{sa}} \oplus \overline{\text{span}_{\mathbb{R}} S}.$$

□

Proposition 6.4 (Strong closure for finite-layer backgrounds). *Let Ω be a finite-layer certified background with a fixed finite certificate. There is a closure operation*

$$\text{cl}_{\Omega} : \mathcal{P}(\Omega) \rightarrow \mathcal{P}(\Omega)$$

such that:

- (i) $S \subset \text{cl}_{\Omega}(S)$;
- (ii) $\text{cl}_{\Omega}(\text{cl}_{\Omega}(S)) = \text{cl}_{\Omega}(S)$;
- (iii) $|\text{cl}_{\Omega}(S)| \leq |S| + \aleph_0$;
- (iv) $S_1 \subset S_2$ implies $\text{cl}_{\Omega}(S_1) \subset \text{cl}_{\Omega}(S_2)$;
- (v) for every increasing chain $(S_i)_{i \in I}$,

$$\text{cl}_{\Omega}\left(\bigcup_i S_i\right) = \bigcup_i \text{cl}_{\Omega}(S_i);$$

- (vi) if $S = \text{cl}_{\Omega}(S)$, then S is itself a finite-layer certified background. Consequently $P_S(x) \in M_{\text{sa}}$ for every $x = x^* \in M$.

Moreover the closure operations may be chosen compatibly with adding one more layer: if U is a new layer over a cl_Ω -closed certified subbackground $\Theta \subset \Omega$ and $\Omega^+ = \Omega \dot{\cup} U$, then cl_{Ω^+} can be chosen so that every cl_{Ω^+} -closed $\Xi \subset \Omega^+$ satisfies

$$\Xi \cap \Omega = \text{cl}_\Omega(\Xi \cap \Omega).$$

Proof. We prove the closure assertions and the compatibility assertion simultaneously by induction on the complexity of the fixed finite certificate. The empty case is trivial. Suppose $\Omega = \Omega^- \dot{\cup} U$, where Ω^- has already been given a closure operation with the stated properties, and where U is a layer over a certified subbackground $\Theta \subset \Omega^-$. Thus there are von Neumann algebras $C \subset R \subset M$ with

$$L^2(R)_{\text{sa}} = L^2(C)_{\text{sa}} \oplus \overline{\text{span}}_{\mathbb{R}}(\Theta \cup U).$$

Choose a large $H(\chi)$ and a countable Skolemized language, with constants for all natural numbers. As in Lemma 6.3, all von Neumann algebras, orthonormal families, conditional expectations, support maps, and previously chosen Skolem functions are named as single predicates, function symbols, or set-valued parameters; we do not name all their individual elements as constants. The language contains:

- all data and Skolem functions used to define the already constructed closure operations for Ω^- and for every certified subbackground appearing in the finite certificate, in particular for Θ ;
- the completed-block data $(C, R, \Theta \cup U)$ and its support function as in Lemma 6.3;
- the algebraic operations on M , constants for all natural numbers, and the relevant conditional expectations.

Since the fixed certificate has finite complexity, only finitely many previously constructed closure languages occur here; their union is still a countable language.

For $S \subset \Omega$, let $\text{Hull}_\Omega(S)$ be the hull of $S \cup \{\omega\}$ together with the structural parameters in this structure, and define

$$\text{cl}_\Omega(S) = \Omega \cap \text{Hull}_\Omega(S).$$

Lemma 6.2 gives the size, monotonicity, and chain-continuity assertions, and the same hull calculation as in Lemma 6.3 gives idempotence.

Assume $S = \text{cl}_\Omega(S)$ and put $\mathcal{E} = \text{Hull}_\Omega(S)$. Then $S = \Omega \cap \mathcal{E}$. Let

$$S^- = S \cap \Omega^-, \quad U_S = S \cap U, \quad \Theta_S = \Theta \cap \mathcal{E}.$$

Because the Skolem functions defining cl_{Ω^-} are included in the present language, S^- is cl_{Ω^-} -closed. Indeed,

$$\text{cl}_{\Omega^-}(S^-) \subset \Omega^- \cap \mathcal{E} = S^-,$$

and the reverse inclusion follows from the definition of a closure operation. Hence S^- is certified by the induction hypothesis. Let cl_Θ be a strong closure operation for the certified background Θ , supplied by the induction hypothesis for the finite certificate of Θ . The Skolem functions defining cl_Θ have been included in the present language. Therefore

$$\text{cl}_\Theta(\Theta_S) \subset \Theta \cap \mathcal{E} = \Theta_S,$$

and the reverse inclusion follows from extensivity of the closure operation. Thus Θ_S is cl_Θ -closed, and hence is a certified subbackground. Since $\Theta \subset \Omega^-$, we also have $\Theta_S \subset S^-$. Because the Skolem functions for the completed block $\mathfrak{B} = (C, R, \Theta \cup U)$ are included in the present language,

$$T_S = (\Theta \cup U) \cap \mathcal{E} = \Theta_S \cup U_S$$

is closed for the block closure from Lemma 6.3. Indeed,

$$\text{cl}_{\mathfrak{B}}(T_S) \subset (\Theta \cup U) \cap \mathcal{E} = T_S,$$

and the reverse inclusion is automatic. Applying that lemma to this closed set gives von Neumann algebras $C_S \subset C$ and $R_S \subset R$ such that

$$L^2(R_S)_{\text{sa}} = L^2(C_S)_{\text{sa}} \oplus \overline{\text{span}}_{\mathbb{R}}(\Theta_S \cup U_S).$$

Thus U_S is a layer over the certified subbackground $\Theta_S \subset S^-$. It follows that

$$S = S^- \dot{\cup} U_S$$

is finite-layer certified.

For the final compatibility assertion, construct cl_{Ω^+} by using a Skolem language which contains all Skolem functions used for cl_{Ω} and the completed-block data defining the new layer U over Θ . Let $\Xi = \text{cl}_{\Omega^+}(\Xi)$ and $\mathcal{E} = \text{Hull}_{\Omega^+}(\Xi)$. Then

$$\Xi = \Omega^+ \cap \mathcal{E}, \quad \Xi \cap \Omega = \Omega \cap \mathcal{E}.$$

Since $\mathcal{E}$ is closed under the finitary Skolem functions defining cl_{Ω} ,

$$\text{cl}_{\Omega}(\Xi \cap \Omega) \subset \Omega \cap \mathcal{E} = \Xi \cap \Omega.$$

The reverse inclusion follows from extensivity, and therefore

$$\Xi \cap \Omega = \text{cl}_{\Omega}(\Xi \cap \Omega).$$

We also need the old support of the new layer to be certified. Since $\Theta = \text{cl}_{\Omega}(\Theta)$ and $\mathcal{E}$ is closed under the Skolem functions defining cl_{Ω} , we have

$$\text{cl}_{\Omega}(\Theta \cap \mathcal{E}) \subset \text{cl}_{\Omega}(\Theta) \cap \mathcal{E} = \Theta \cap \mathcal{E}.$$

Again the reverse inclusion follows from extensivity, so

$$\Theta \cap \mathcal{E} = \text{cl}_{\Omega}(\Theta \cap \mathcal{E}).$$

Thus $\Theta \cap \mathcal{E}$ is a certified subbackground by the closure assertions already proved for Ω .

Finally, apply Lemma 6.3 to the completed block $(C, R, \Theta \cup U)$. Since the corresponding block-closure Skolem functions are included in the language,

$$(\Theta \cup U) \cap \mathcal{E} = (\Theta \cap \mathcal{E}) \cup (\Xi \cap U)$$

is closed for that block closure. Hence there are von Neumann algebras $C_{\Xi} \subset C$ and $R_{\Xi} \subset R$ such that

$$L^2(R_{\Xi})_{\text{sa}} = L^2(C_{\Xi})_{\text{sa}} \oplus \overline{\text{span}}_{\mathbb{R}}((\Theta \cap \mathcal{E}) \cup (\Xi \cap U)).$$

Thus $\Xi \cap U$ is a certified layer over the certified subbackground $\Theta \cap \mathcal{E}$. Since

$$\Theta \cap \mathcal{E} \subset \Xi \cap \Omega$$

and $\Xi \cap \Omega$ is cl_{Ω} -closed, the closure assertions already proved for Ω show that $\Xi \cap \Omega$ is finite-layer certified. Therefore

$$\Xi = (\Xi \cap \Omega) \dot{\cup} (\Xi \cap U)$$

is finite-layer certified by adjoining the layer $\Xi \cap U$ over the certified subbackground $\Theta \cap \mathcal{E} \subset \Xi \cap \Omega$. $\square$

Corollary 6.5 (Restriction and adjoining a layer). *Let $(\Omega, \mathfrak{c}_{\Omega})$ be a finite-layer certified background, and let cl_{Ω} be a closure operation supplied by Proposition 6.4.*

- (i) *If $S \subset \Omega$ satisfies $S = \text{cl}_{\Omega}(S)$, then S admits a finite-layer certificate $\mathfrak{c}_S$. With this certificate, $P_{\mathfrak{c}_S}(x)$ is the Hilbert-space projection of x onto $\overline{\text{span}}_{\mathbb{R}} S$ for every $x = x^* \in M$.*

- (ii) Suppose $S = \text{cl}_\Omega(S)$ and $U \subset \mathcal{S}_0(M)$ is an orthonormal family orthogonal to Ω . Suppose further that there are von Neumann algebras $C \subset R \subset M$ such that

$$L^2(R)_{\text{sa}} = L^2(C)_{\text{sa}} \oplus \overline{\text{span}}_{\mathbb{R}}(S \cup U).$$

Then $\Omega \dot{\cup} U$ is a finite-layer certified background. A certificate is obtained by taking the old certificate $\mathfrak{c}_\Omega$, the certificate $\mathfrak{c}_S$ from part (i), and adjoining the single layer U over S .

- (iii) The strong closure operation on $\Omega \dot{\cup} U$ may be chosen compatibly with cl_Ω in the sense of Proposition 6.4.

Proof. Part (i) is exactly Proposition 6.4(vi), together with Lemma 5.5. Part (ii) follows directly from the recursive definition of finite-layer certificates in Definition 5.3. Part (iii) is the compatibility assertion in Proposition 6.4. $\square$

7. THE COUNTABLE CERTIFIED EXTENSION

We first prove the extension theorem in the countable case. The proof uses a standard fair bookkeeping device.

Lemma 7.1 (Fair bookkeeping with countable batches). *Suppose that a recursion is indexed by $n \geq 1$, and that at each stage at most countably many new tasks may appear. Then the tasks can be labelled and scheduled so that every task which eventually appears is processed infinitely often at times n_k satisfying*

$$\sum_k \frac{1}{n_k} = \infty.$$

Proof. For $j \in \{0, 1, 2, \dots\}$, put

$$A_j = \{n \geq 1 : \nu_2(n) = j\},$$

where $\nu_2(n)$ denotes the largest ν such that $2^\nu \mid n$. Then the sets A_j partition the positive integers, and

$$\sum_{n \in A_j} \frac{1}{n} = \infty \quad (j = 0, 1, 2, \dots).$$

Next partition the label set $\{0, 1, 2, \dots\}$ into pairwise disjoint infinite subsets

$$\{0, 1, 2, \dots\} = \bigsqcup_{m=0}^{\infty} L_m.$$

The labels in L_m will be reserved for tasks which first appear at stage m .

At stage m , the new batch of tasks is countable, so we assign those tasks injectively to unused labels in the infinite set L_m . At time $n \geq 1$, let $j = \nu_2(n)$. If a task has already been assigned label j , we process that task; otherwise we do nothing at time n .

If a task first appears at stage m and receives label $j \in L_m$, then it is processed at all sufficiently late times in A_j . Removing finitely many early terms from A_j does not change the divergence of the harmonic sum. Hence the visiting times n_k of this task satisfy

$$\sum_k \frac{1}{n_k} = \infty.$$

$\square$

Lemma 7.2 (Countable certified extension). *Let $\kappa = \text{dens } L^2(M) > \aleph_0$. Let $N \subset M$ be a von Neumann subalgebra with $\text{dens } L^2(N) < \kappa$. Let $\Omega \subset \mathcal{S}_0(M)$ be a finite-layer certified background with $|\Omega| < \kappa$, $E_N(\Omega) = 0$, and a fixed strong closure operation cl_Ω as in Proposition 6.4. Let $S \subset \Omega$, $X \subset M_{\text{sa}}$, and $Y \subset N_{\text{sa}}$ be countable. Then there are von Neumann algebras $C \subset N$ and $C \subset R \subset M$, a cl_Ω -closed set $\Omega_0 \subset \Omega$, and an orthonormal family $U \subset \mathcal{S}_0(M)$ such that*

$$\begin{aligned} S &\subset \Omega_0, & X &\subset R, & Y &\subset C, \\ |\Omega_0|, |U|, \text{dens } L^2(R) &\leq \aleph_0, \\ E_N(U) &= 0, & U &\perp \Omega, \end{aligned}$$

and

$$L^2(R)_{\text{sa}} = L^2(C)_{\text{sa}} \oplus \overline{\text{span}}_{\mathbb{R}}(\Omega_0 \cup U).$$

Consequently U is a layer over Ω_0 .

Proof. We run a countable Henkin construction. Maintain countable sets of parameters $\mathcal{C} \subset N_{\text{sa}}$, $\Omega_* \subset \Omega$, and a finite or countable orthonormal family of already chosen symmetries U_* . Initially

$$\mathcal{C} = Y, \quad \Omega_* = \text{cl}_\Omega(S), \quad U_* = \emptyset.$$

We implement the task list in countable batches, using Lemma 7.1.

By a formal $*$ -polynomial word we mean a noncommutative $*$ -polynomial with coefficients in $\mathbb{Q} + i\mathbb{Q}$ in finitely many parameters which have appeared by the relevant stage. For such a word w , we use the self-adjoint parts

$$\Re w = \frac{w + w^*}{2}, \quad \Im w = \frac{w - w^*}{2i}.$$

An *eventual self-adjoint word* is an evaluated element of M_{sa} which is equal to $\Re w$ or $\Im w$ for some such formal word w , after all parameters occurring in w have appeared at some finite stage of the construction.

At stage 0, declare all elements of the initial parameter set

$$\mathcal{C} \cup X \cup \Omega_* \cup U_*$$

to have appeared. Enumerate all formal $*$ -polynomial words with coefficients in $\mathbb{Q} + i\mathbb{Q}$ using these appeared parameters, take their real and imaginary self-adjoint parts, and regard the resulting self-adjoint words as the first countable batch of tasks.

At the end of each later stage, after $\mathcal{C}$, Ω_* , or U_* has possibly been enlarged, declare all newly added parameters to have appeared. Then enumerate all formal $*$ -polynomial words with coefficients in $\mathbb{Q} + i\mathbb{Q}$ using the appeared parameters, take their real and imaginary self-adjoint parts, and add precisely those self-adjoint words which have not previously been labelled as the next countable batch of tasks.

By Lemma 7.1, the labels can be assigned so that every eventual self-adjoint word is processed infinitely many times at visiting times n_k satisfying

$$\sum_k \frac{1}{n_k} = \infty.$$

At a stage whose scheduled label has not yet been assigned, we perform no processing and continue to the next stage. Otherwise the scheduled label corresponds to a bounded self-adjoint word over the appeared parameter set, and we process that word as follows.

Thus suppose that the current scheduled task is represented by a bounded self-adjoint word $y \in M_{\text{sa}}$. Define

$$b(y) = y - E_N(y) - P_\Omega(y).$$

This is a bounded self-adjoint element of M : indeed, $E_N(y) \in N_{\text{sa}} \subset M_{\text{sa}}$, and $P_\Omega(y) \in M_{\text{sa}}$ by Lemma 5.5.

Moreover,

$$E_N(P_\Omega(y)) = 0,$$

because $P_\Omega(y)$ is an L^2 -limit of finite real linear combinations of elements of Ω , while E_N is L^2 -continuous and vanishes on Ω . Hence

$$E_N(b(y)) = 0.$$

Also

$$b(y) \perp \overline{\text{span}}_{\mathbb{R}} \Omega.$$

Indeed, $y - P_\Omega(y)$ is orthogonal to $\overline{\text{span}}_{\mathbb{R}} \Omega$ by definition of P_Ω , and $E_N(y) \perp \Omega$ because $E_N(y) \in N_{\text{sa}}$ and $E_N(\Omega) = 0$.

Add $E_N(y)$ to $\mathcal{C}$, add the countable set $\text{supp}_\Omega(P_\Omega(y))$ to Ω_* , and then replace Ω_* by $\text{cl}_\Omega(\Omega_*)$. This keeps Ω_* countable by Proposition 6.4.

At this finite stage, only finitely many symmetries have been chosen; write them as $u_1, \dots, u_m$. Since $b(y)$ and the u_i 's are self-adjoint, the real Hilbert-space projection is

$$P_{\text{span}_{\mathbb{R}}\{u_1, \dots, u_m\}} b(y) = \sum_{i=1}^m \langle b(y), u_i \rangle_2 u_i,$$

with real coefficients, and hence belongs to M_{sa} . Put

$$r = b(y) - P_{\text{span}_{\mathbb{R}}\{u_1, \dots, u_m\}} b(y).$$

Thus $r \in M_{\text{sa}}$. We record explicitly that r satisfies the hypotheses needed for Lemma 4.4. Since $E_N(b(y)) = 0$ and $E_N(u_i) = 0$ for $1 \leq i \leq m$, we have

$$E_N(r) = 0.$$

Moreover $b(y) \perp \Omega$ and $u_i \perp \Omega$ for all i , hence

$$\tau(\omega r) = 0 \quad (\omega \in \Omega).$$

By construction r is the residual after orthogonal projection onto $\text{span}_{\mathbb{R}}\{u_1, \dots, u_m\}$, so

$$\tau(u_i r) = 0 \quad (1 \leq i \leq m).$$

Therefore r is orthogonal to

$$\Gamma = \Omega \cup \{u_1, \dots, u_m\}.$$

If $r = 0$, choose no new symmetry at this stage. If $r \neq 0$, apply Lemma 4.4 with this constraint set Γ . This set has cardinal $< \kappa$. We obtain $u_{m+1} \in \mathcal{S}_0(M)$ satisfying

$$E_N(u_{m+1}) = 0, \quad u_{m+1} \perp \Omega, \quad u_{m+1} \perp u_i \quad (1 \leq i \leq m),$$

and

$$\langle r, u_{m+1} \rangle_2 = \frac{\|r\|_2^2}{\|r\|_\infty}.$$

Add u_{m+1} to U_* .

Let U be the family of all chosen symmetries, let Ω_0 be the union of the increasing sequence of values of Ω_* , let $C = W^*(\mathcal{C})$, and put

$$R = W^*(C, X, \Omega_0, U).$$

The sets Ω_0 and U are countable. Moreover, R is generated by countably many bounded elements. Hence, by Lemma 4.1(i),

$$\text{dens } L^2(R) \leq \aleph_0.$$

The set Ω_0 is cl_Ω -closed by the continuity of the closure operation along the increasing sequence of values of Ω_* .

It remains to show that the announced decomposition holds. Fix an eventual self-adjoint word y . Once all parameters occurring in y have appeared, the word y receives a fixed label.

For $n \geq 1$, let $V_n(y)$ denote the real span of the symmetries which have been chosen before stage n . Equivalently, if m_n symmetries have been chosen before stage n , then

$$V_n(y) = \text{span}_{\mathbb{R}}\{u_1, \dots, u_{m_n}\}.$$

Put

$$D_n(y) = \text{dist}(b(y), V_n(y)).$$

The sequence $D_n(y)$ is decreasing, and its limit is

$$d = \text{dist}(b(y), \overline{\text{span}_{\mathbb{R}} U}).$$

We prove that $d = 0$.

Suppose, toward a contradiction, that $d > 0$. Let

$$n_1 < n_2 < \dots$$

be the visiting times at which the fixed label of y is processed after y has appeared. By Lemma 7.1,

$$\sum_k \frac{1}{n_k} = \infty.$$

At stage n_k , the current residual is

$$r_k = b(y) - P_{V_{n_k}(y)} b(y).$$

Hence

$$\|r_k\|_2 = D_{n_k}(y) \geq d.$$

In particular $r_k \neq 0$, so the construction chooses a new symmetry at stage n_k .

Since at most one new symmetry is chosen at each stage, we have $m_{n_k} \leq n_k$. Also,

$$\begin{aligned} \|r_k\|_{\infty} &\leq \|b(y)\|_{\infty} + \|P_{V_{n_k}(y)} b(y)\|_{\infty} \\ &\leq \|b(y)\|_{\infty} + \sum_{i=1}^{m_{n_k}} |\langle b(y), u_i \rangle_2| \\ &\leq \|b(y)\|_{\infty} + \sqrt{m_{n_k}} \|b(y)\|_2 \\ &\leq K_y \sqrt{n_k}, \end{aligned}$$

where

$$K_y = \|b(y)\|_{\infty} + \|b(y)\|_2.$$

If $K_y = 0$, then $b(y) = 0$ and there is nothing to prove. Thus we may assume $K_y > 0$.

At stage n_k , let u be the new symmetry chosen by Lemma 4.4. Then $u \perp V_{n_k}(y)$, and

$$\langle r_k, u \rangle_2 = \frac{\|r_k\|_2^2}{\|r_k\|_{\infty}}.$$

Therefore

$$\begin{aligned} D_{n_k+1}(y)^2 &\leq \text{dist}(b(y), V_{n_k}(y) + \mathbb{R}u)^2 \\ &= \|r_k\|_2^2 - |\langle r_k, u \rangle_2|^2 \\ &= D_{n_k}(y)^2 - \frac{\|r_k\|_2^4}{\|r_k\|_{\infty}^2}. \end{aligned}$$

Since $D_n(y)$ is decreasing, summing over $k = 1, \dots, \ell$ gives

$$\sum_{k=1}^{\ell} \frac{\|r_k\|_2^4}{\|r_k\|_{\infty}^2} \leq D_{n_1}(y)^2.$$

Using $\|r_k\|_2 \geq d$ and $\|r_k\|_{\infty} \leq K_y \sqrt{n_k}$, we get

$$\sum_{k=1}^{\ell} \frac{d^4}{K_y^2 n_k} \leq D_{n_1}(y)^2.$$

Letting $\ell \rightarrow \infty$ contradicts

$$\sum_k \frac{1}{n_k} = \infty.$$

Therefore $d = 0$, and hence

$$b(y) \in \overline{\text{span}_{\mathbb{R}} U}$$

for every eventual self-adjoint word y .

For such y we have

$$y = E_N(y) + P_{\Omega}(y) + b(y),$$

where $E_N(y) \in L^2(C)_{\text{sa}}$, $P_{\Omega}(y) \in \overline{\text{span}_{\mathbb{R}} \Omega_0}$ by the support bookkeeping, and $b(y) \in \overline{\text{span}_{\mathbb{R}} U}$. The eventual words are L^2 -dense in $L^2(R)_{\text{sa}}$. Let $\mathcal{C}_{\infty}$ be the final value of $\mathcal{C}$, and put

$$\mathcal{P}_{\infty} = \mathcal{C}_{\infty} \cup X \cup \Omega_0 \cup U.$$

Then

$$C = W^*(\mathcal{C}_{\infty}), \quad R = W^*(C, X, \Omega_0, U) = W^*(\mathcal{P}_{\infty}).$$

Every element of $\mathcal{P}_{\infty}$ appears at some finite stage of the construction. Hence every unital rational-coefficient $*$ -word in $\mathcal{P}_{\infty}$, and its real and imaginary self-adjoint parts, is an eventual word. Let

$$\mathcal{A}_{\mathbb{Q}} = *-\text{alg}_{\mathbb{Q}+i\mathbb{Q}}(\mathcal{P}_{\infty}).$$

The self-adjoint part of $\mathcal{A}_{\mathbb{Q}}$ is contained in the set of eventual self-adjoint words. Its norm closure is the C^* -algebra generated by $\mathcal{P}_{\infty}$, whose bicommutant is R . By Kaplansky density, $(\mathcal{A}_{\mathbb{Q}})_{\text{sa}}$ is L^2 -dense in $L^2(R)_{\text{sa}}$. Therefore the eventual self-adjoint words are L^2 -dense in $L^2(R)_{\text{sa}}$.

The right-hand side below is closed, since $L^2(C)_{\text{sa}}$ is orthogonal to $\overline{\text{span}_{\mathbb{R}}(\Omega_0 \cup U)}$. This orthogonality follows from $C \subset N$, $E_N(\Omega_0) = 0$, $E_N(U) = 0$, and $U \perp \Omega$. Hence

$$L^2(R)_{\text{sa}} \subset L^2(C)_{\text{sa}} + \overline{\text{span}_{\mathbb{R}}(\Omega_0 \cup U)}.$$

The reverse inclusion is clear from the definition of R . This proves the completed-block decomposition. $\square$

8. THE ALL-CARDINAL CERTIFIED EXTENSION THEOREM

We first record the elementary continuity fact used at limit stages.

Lemma 8.1 (Continuity of nested completed blocks). *Let $(C_i, R_i, T_i)_{i \in I}$ be an increasing chain of completed blocks, in the sense that $C_i \subset C_j$, $R_i \subset R_j$, and $T_i \subset T_j$ whenever $i \leq j$. Put*

$$C = W^*\left(\bigcup_i C_i\right), \quad R = W^*\left(\bigcup_i R_i\right), \quad T = \bigcup_i T_i.$$

Then

$$L^2(R)_{\text{sa}} = L^2(C)_{\text{sa}} \oplus \overline{\text{span}_{\mathbb{R}} T}.$$

Proof. In a finite von Neumann algebra, if $R = W^*(\bigcup_i R_i)$ for an increasing family, then

$$L^2(R) = \overline{\bigcup_i L^2(R_i)}^{\|\cdot\|_2}.$$

Indeed, the strong-* closure of the algebraic union is R , and bounded strong-* convergence implies L^2 -convergence; Kaplansky density then gives the equality. The same holds for C .

The subspace $L^2(C)_{\text{sa}}$ is orthogonal to $\overline{\text{span}}_{\mathbb{R}} T$ because this orthogonality holds at every stage and passes to L^2 -closures. The closed sum contains each $L^2(R_i)_{\text{sa}}$, hence contains the L^2 -closure of their union, which is $L^2(R)_{\text{sa}}$. The reverse inclusion is immediate from $C \subset R$ and $T \subset R$. $\square$

Remark 8.2. No regularity assumption on κ is used. We identify cardinals with initial ordinals. If $\lambda > \aleph_0$ is an initial cardinal and $\alpha < \lambda$, then $|\alpha| < \lambda$, and hence

$$\max\{\aleph_0, |\alpha|\} < \lambda.$$

The case $\lambda = \aleph_0$ is handled separately as the base case in Proposition 8.3; in that case the displayed strict inequality with λ on the right is not used.

In all applications below, whenever $\lambda < \kappa$ and $\alpha < \lambda$, one still has

$$\max\{\aleph_0, |\alpha|\} < \kappa.$$

Indeed, if $\lambda = \aleph_0$, then the left-hand side is $\aleph_0 < \kappa$; if $\lambda > \aleph_0$, then the preceding paragraph gives $\max\{\aleph_0, |\alpha|\} < \lambda < \kappa$. Thus the argument does not use regularity of κ , and this remains true when κ is singular.

Proposition 8.3 (Certified extension theorem). *Let $\kappa = \text{dens } L^2(M) > \aleph_0$ and let $\aleph_0 \leq \lambda < \kappa$. Let $N \subset M$ satisfy $\text{dens } L^2(N) < \kappa$. Let $\Omega \subset \mathcal{S}_0(M)$ be a finite-layer certified background with $|\Omega| < \kappa$, $E_N(\Omega) = 0$, and a fixed strong closure operation cl_Ω as in Proposition 6.4. Let*

$$S \subset \Omega, \quad X \subset M_{\text{sa}}, \quad Y \subset N_{\text{sa}}$$

with $|S|, |X|, |Y| \leq \lambda$. Then there are von Neumann algebras $C \subset N$ and $C \subset R \subset M$, a cl_Ω -closed set $\Omega_0 \subset \Omega$, and an orthonormal family $U \subset \mathcal{S}_0(M)$ such that

$$\begin{aligned} S &\subset \Omega_0, & X &\subset R, & Y &\subset C, \\ |\Omega_0|, |U|, \text{dens } L^2(R) &\leq \lambda, \\ E_N(U) &= 0, & U &\perp \Omega, \end{aligned}$$

and

$$L^2(R)_{\text{sa}} = L^2(C)_{\text{sa}} \oplus \overline{\text{span}}_{\mathbb{R}}(\Omega_0 \cup U).$$

Thus U is a layer over Ω_0 .

Proof. We prove the proposition by induction on λ . The case $\lambda = \aleph_0$ is Lemma 7.2.

Assume $\lambda > \aleph_0$ and that the proposition has been proved for all cardinals μ with $\aleph_0 \leq \mu < \lambda$. Enumerate the disjoint union of the data S , X , and Y as $(z_\alpha)_{\alpha < \lambda}$, with repetitions or dummy entries if necessary, remembering the type of each genuine datum. We first apply the countable case to the original background Ω , with support $\text{cl}_\Omega(\emptyset)$ and with $X = Y = \emptyset$. This gives an initial completed state

$$(C_0, R_0, \Omega_0 \cup U_0)$$

with countable density, where Ω_0 is cl_Ω -closed and U_0 is a layer over Ω_0 . The recursion below starts from this state; at successor stages we enlarge the current state to absorb the next datum.

We recursively construct, for $\alpha < \lambda$, completed states

$$(C_\alpha, R_\alpha, \Omega_\alpha \cup U_\alpha)$$

such that

$$C_\alpha \subset N, \quad C_\alpha \subset R_\alpha \subset M, \quad \Omega_\alpha \subset \Omega, \quad \Omega_\alpha = \text{cl}_\Omega(\Omega_\alpha),$$

U_α is a layer over Ω_α ,

$$E_N(U_\alpha) = 0, \quad U_\alpha \perp \Omega,$$

and

$$\text{dens } L^2(R_\alpha), |\Omega_\alpha|, |U_\alpha| \leq \max\{\aleph_0, |\alpha|\}.$$

The states are required to be increasing in all coordinates.

Suppose the state at α is constructed. Put

$$\lambda_\alpha = \max\{\aleph_0, |\alpha|\} < \lambda.$$

Here λ is identified with its initial ordinal. No regularity of λ , and hence no regularity of κ , is used in this inequality; see Remark 8.2. By the induction invariant,

$$\Omega_\alpha = \text{cl}_\Omega(\Omega_\alpha)$$

and

$$L^2(R_\alpha)_{\text{sa}} = L^2(C_\alpha)_{\text{sa}} \oplus \overline{\text{span}_\mathbb{R}}(\Omega_\alpha \cup U_\alpha).$$

Moreover U_α is orthonormal and orthogonal to Ω . Therefore Corollary 6.5, applied with $S = \Omega_\alpha$ and $U = U_\alpha$, certifies the background

$$\Omega \dot{\cup} U_\alpha.$$

This is a re-certification of the whole currently constructed family U_α as a single layer over Ω_α ; the finite certificate depth is not accumulated along the transfinite recursion. The same corollary supplies a strong closure operation on $\Omega \dot{\cup} U_\alpha$ compatible with the original operation cl_Ω . Also

$$E_N(\Omega \dot{\cup} U_\alpha) = 0$$

and

$$|\Omega \dot{\cup} U_\alpha| < \kappa,$$

because $|\Omega| < \kappa$ and $|U_\alpha| \leq \lambda_\alpha < \lambda < \kappa$.

Choose self-adjoint L^2 -dense sets

$$Y_\alpha \subset (C_\alpha)_{\text{sa}}, \quad X_\alpha \subset (R_\alpha)_{\text{sa}},$$

of cardinality at most λ_α . This is possible because

$$\text{dens } L^2(C_\alpha) \leq \text{dens } L^2(R_\alpha) \leq \lambda_\alpha$$

and the bounded self-adjoint parts are L^2 -dense in the corresponding self-adjoint L^2 -spaces. By Lemma 4.1(iv),

$$W^*(Y_\alpha) = C_\alpha, \quad W^*(X_\alpha) = R_\alpha.$$

Set

$$S'_\alpha = \Omega_\alpha \cup U_\alpha, \quad X'_\alpha = X_\alpha, \quad Y'_\alpha = Y_\alpha.$$

If the datum z_α comes from the original set S , add it to S'_α ; if it comes from X , add it to X'_α ; and if it comes from Y , add it to Y'_α . Apply the induction hypothesis at cardinal λ_α to the background $\Omega \dot{\cup} U_\alpha$ with input $S'_\alpha, X'_\alpha, Y'_\alpha$. We obtain a completed extension

$$(C_{\alpha+1}, R_{\alpha+1}, \Xi_{\alpha+1} \cup V_{\alpha+1}),$$

where

$$\Xi_{\alpha+1} \subset \Omega \dot{\cup} U_\alpha$$

is closed for the chosen closure operation on $\Omega \dot{\cup} U_\alpha$ and contains $\Omega_\alpha \cup U_\alpha$. Since all of U_α is required support,

$$\Xi_{\alpha+1} = (\Xi_{\alpha+1} \cap \Omega) \dot{\cup} U_\alpha.$$

Set

$$\Omega_{\alpha+1} = \Xi_{\alpha+1} \cap \Omega, \quad U_{\alpha+1} = U_\alpha \cup V_{\alpha+1}.$$

By the compatibility part of Proposition 6.4, $\Omega_{\alpha+1}$ is cl_Ω -closed. Moreover the new state contains the old one. Indeed, $Y_\alpha \subset C_{\alpha+1}$ and $X_\alpha \subset R_{\alpha+1}$ by the choice of the input data for the induction hypothesis; since

$$W^*(Y_\alpha) = C_\alpha, \quad W^*(X_\alpha) = R_\alpha,$$

we get

$$C_\alpha \subset C_{\alpha+1}, \quad R_\alpha \subset R_{\alpha+1}.$$

Also $\Omega_\alpha \subset \Omega_{\alpha+1}$ and $U_\alpha \subset U_{\alpha+1}$ by construction. Thus the completed states are increasing in all coordinates.

The completed-block decomposition for the output says exactly that

$$L^2(R_{\alpha+1})_{\text{sa}} = L^2(C_{\alpha+1})_{\text{sa}} \oplus \overline{\text{span}}_{\mathbb{R}}(\Omega_{\alpha+1} \cup U_{\alpha+1}).$$

The cardinal bounds are $\leq \lambda_\alpha \leq \max\{\aleph_0, |\alpha + 1|\}$. Thus the successor step is complete.

At a limit ordinal $\delta < \lambda$, define

$$\begin{aligned} C_\delta &= W^*\left(\bigcup_{\alpha < \delta} C_\alpha\right), & R_\delta &= W^*\left(\bigcup_{\alpha < \delta} R_\alpha\right), \\ \Omega_\delta &= \bigcup_{\alpha < \delta} \Omega_\alpha, & U_\delta &= \bigcup_{\alpha < \delta} U_\alpha. \end{aligned}$$

Since $\delta < \lambda$ and λ is an initial cardinal, $|\delta| < \lambda$. The cardinal estimates are as follows. For each $\alpha < \delta$, choose an L^2 -dense subset $D_\alpha \subset L^2(R_\alpha)$ of cardinal at most $\max\{\aleph_0, |\alpha|\}$. Then

$$\left| \bigcup_{\alpha < \delta} D_\alpha \right| \leq |\delta| \cdot \max\{\aleph_0, |\delta|\} = \max\{\aleph_0, |\delta|\}.$$

By Lemma 4.1(ii), this union is L^2 -dense in $L^2(R_\delta)$. The same cardinal calculation gives

$$|\Omega_\delta|, |U_\delta| \leq \max\{\aleph_0, |\delta|\}.$$

Hence

$$\text{dens } L^2(R_\delta), |\Omega_\delta|, |U_\delta| \leq \max\{\aleph_0, |\delta|\} < \lambda.$$

The continuity of cl_Ω from Proposition 6.4 shows that Ω_δ is cl_Ω -closed. Lemma 8.1 applied to the nested completed blocks gives

$$L^2(R_\delta)_{\text{sa}} = L^2(C_\delta)_{\text{sa}} \oplus \overline{\text{span}}_{\mathbb{R}}(\Omega_\delta \cup U_\delta).$$

Thus U_δ is a layer over Ω_δ .

Finally put

$$\begin{aligned} C &= W^*\left(\bigcup_{\alpha < \lambda} C_\alpha\right), & R &= W^*\left(\bigcup_{\alpha < \lambda} R_\alpha\right), \\ \Omega_{\text{out}} &= \bigcup_{\alpha < \lambda} \Omega_\alpha, & U &= \bigcup_{\alpha < \lambda} U_\alpha. \end{aligned}$$

The continuity of cl_Ω gives that Ω_{out} is cl_Ω -closed, and Lemma 8.1 gives

$$L^2(R)_{\text{sa}} = L^2(C)_{\text{sa}} \oplus \overline{\text{span}}_{\mathbb{R}}(\Omega_{\text{out}} \cup U).$$

All data from S , X , and Y were absorbed during the recursion. The cardinal bounds for Ω_{out} and U follow from taking a union of λ many sets each of cardinal at most λ :

$$|\Omega_{\text{out}}|, |U| \leq \lambda.$$

It remains only to record the density bound for R . For each $\alpha < \lambda$, choose an L^2 -dense set $D_\alpha \subset L^2(R_\alpha)$ with $|D_\alpha| \leq \lambda$. Then

$$\left| \bigcup_{\alpha < \lambda} D_\alpha \right| \leq \lambda \cdot \lambda = \lambda.$$

By Lemma 4.1(ii), the union $\bigcup_{\alpha < \lambda} D_\alpha$ is L^2 -dense in $L^2(R)$. Hence

$$\text{dens } L^2(R) \leq \lambda.$$

Thus the conclusion holds with $\Omega_0 = \Omega_{\text{out}}$. $\square$

9. RELATIVE EXTENSION AND THE NONSEPARABLE THEOREM

Proposition 9.1 (Relative extension property). *Let $\kappa = \text{dens } L^2(M) > \aleph_0$ and let $\aleph_0 \leq \lambda < \kappa$. Let $N \subset M$ be a von Neumann subalgebra with $\text{dens } L^2(N) \leq \lambda$. Suppose $H_0(N)$ has an orthonormal basis $B_N \subset \mathcal{S}_0(N)$. If $X \subset M_{\text{sa}}$ and $|X| \leq \lambda$, then there are a von Neumann algebra P with $N \cup X \subset P \subset M$ and an orthonormal family $U \subset \mathcal{S}_0(P)$ such that*

$$\text{dens } L^2(P) \leq \lambda,$$

$$U \perp L^2(N)_{\text{sa}},$$

and

$$B_N \cup U$$

is an orthonormal basis of $H_0(P)$.

Proof. Apply Proposition 8.3 with the empty background $\Omega = \emptyset$, $S = \emptyset$, and with $Y \subset N_{\text{sa}}$ a self-adjoint L^2 -dense set of cardinal $\leq \lambda$. We obtain $C \subset N$, $C \subset R \subset M$, and $U \subset \mathcal{S}_0(M)$ such that $Y \subset C$, $X \subset R$, and

$$L^2(R)_{\text{sa}} = L^2(C)_{\text{sa}} \oplus \overline{\text{span}_{\mathbb{R}} U}.$$

Since Y is L^2 -dense in $L^2(N)_{\text{sa}}$, Lemma 4.1(iv) gives $W^*(Y) = N$. Hence $C = N$. Put $P = R$. Then $N = C \subset R = P$ and $X \subset P$, and

$$L^2(P)_{\text{sa}} = L^2(N)_{\text{sa}} \oplus \overline{\text{span}_{\mathbb{R}} U}.$$

In particular,

$$U \perp L^2(N)_{\text{sa}}.$$

Moreover $U \subset P$. Indeed, each $u \in U$ belongs to M , and its L^2 -vector belongs to $L^2(P)$ by the displayed decomposition. Hence Lemma 4.1(iii) gives $u \in P$. Since each $u \in U$ is a trace-zero symmetry in M , it is also a trace-zero symmetry in P . Thus $U \subset \mathcal{S}_0(P)$.

Since $U \subset \mathcal{S}_0(P)$, every element of U is orthogonal to 1. Taking the orthogonal complement of $\mathbb{R}1$ in the displayed decomposition gives

$$H_0(P) = H_0(N) \oplus \overline{\text{span}_{\mathbb{R}} U}.$$

Since B_N is an orthonormal basis of $H_0(N)$, the family $B_N \cup U$ is an orthonormal basis of $H_0(P)$. $\square$

Lemma 9.2 (A separable abelian starting algebra). *Every II_1 factor contains a separable diffuse abelian von Neumann subalgebra A_0 . Moreover $H_0(A_0)$ has an orthonormal basis contained in $\mathcal{S}_0(A_0)$.*

Proof. We construct a dyadic system of projections. Start with $p_\emptyset = 1$. By the comparison and dimension theory of projections in a finite factor [KR97b, Ch. 6], every nonzero projection in M can be split into two orthogonal subprojections of equal trace. Recursively, for every finite binary string $\sigma \in \{0, 1\}^{<\mathbb{N}}$, choose orthogonal projections $p_{\sigma 0}, p_{\sigma 1} \leq p_\sigma$ such that

$$p_{\sigma 0} + p_{\sigma 1} = p_\sigma, \quad \tau(p_{\sigma 0}) = \tau(p_{\sigma 1}) = \frac{1}{2}\tau(p_\sigma).$$

The projections in this dyadic tree commute pairwise, since any two of them are either orthogonal or one is dominated by the other. Let A_0 be the abelian von Neumann algebra generated by all projections p_σ . For $n \geq 0$, put

$$F_n = \text{span}\{p_\sigma : |\sigma| = n\}.$$

Then $F_n \subset F_{n+1}$, and

$$A_0 = W^*\left(\bigcup_{n \geq 0} F_n\right).$$

Hence, by Lemma 4.1(ii),

$$L^2(A_0) = \overline{\bigcup_{n \geq 0} L^2(F_n)}^{\|\cdot\|_2},$$

so $L^2(A_0)$ is separable.

We claim that A_0 is diffuse. Let $0 \neq q \in A_0$ be a projection. Choose n with $2^{-n} < \tau(q)$. Since $\sum_{|\sigma|=n} p_\sigma = 1$, there is some $\sigma \in \{0, 1\}^n$ such that $qp_\sigma \neq 0$. If $qp_\sigma = q$, then $q \leq p_\sigma$, which would imply

$$\tau(q) \leq \tau(p_\sigma) = 2^{-n},$$

a contradiction. Thus

$$0 < qp_\sigma < q.$$

Since A_0 is abelian, qp_σ is a projection in A_0 . Hence q is not minimal, and A_0 is diffuse.

For every finite set $F \subset \mathbb{N}$, choose n with $F \subset \{1, \dots, n\}$, and define

$$w_F = \sum_{\sigma \in \{0, 1\}^n} (-1)^{\sum_{j \in F} \sigma_j} p_\sigma.$$

This definition is independent of the choice of n . Each w_F is a self-adjoint unitary in A_0 , $w_\emptyset = 1$, and $\tau(w_F) = 0$ whenever $F \neq \emptyset$. For each n , the finite family

$$\{w_F : F \subset \{1, \dots, n\}\}$$

is the usual Walsh orthonormal basis of $L^2(F_n)_{\text{sa}}$. Therefore

$$\{w_F : F \subset \mathbb{N}, |F| < \infty\}$$

is an orthonormal basis of $L^2(A_0)_{\text{sa}}$, and the nonconstant Walsh symmetries

$$\{w_F : 0 < |F| < \infty\}$$

form an orthonormal basis of $H_0(A_0)$ contained in $\mathcal{S}_0(A_0)$. □

Theorem 9.3 (Nonseparable case). *Let M be a II_1 factor with faithful normal tracial state τ , and set $\kappa = \text{dens } L^2(M, \tau)$. If*

$$\kappa > \aleph_0,$$

then there is a family $(s_i)_{i \in I}$ of self-adjoint unitaries in M , with $|I| = \kappa$, such that $\{\widehat{s}_i : i \in I\}$ is a Hilbert orthonormal basis of $L^2(M, \tau)$. Consequently $L^2(M, \tau)$ has a Hilbert orthonormal basis consisting of trace vectors for the left action of M .

Proof. Let

$$\kappa = \text{dens } L^2(M, \tau) > \aleph_0.$$

The decompositions

$$L^2(M)_{\text{sa}} = H_0(M) \oplus \mathbb{R}1, \quad L^2(M) = L^2(M)_{\text{sa}} + iL^2(M)_{\text{sa}}$$

show that $\text{dens } H_0(M) = \kappa$. The bounded trace-zero self-adjoint elements $M_{\text{sa}} \cap H_0(M)$ are L^2 -dense in $H_0(M)$. Indeed, if $\xi \in H_0(M)$, then by definition of $L^2(M)_{\text{sa}}$ there are $x_i = x_i^* \in M$ with $\|x_i - \xi\|_2 \rightarrow 0$. Since

$$\tau(x_i) = \langle x_i, 1 \rangle_2 \longrightarrow \langle \xi, 1 \rangle_2 = 0,$$

we have

$$x_i - \tau(x_i)1 \in M_{\text{sa}} \cap H_0(M)$$

and

$$\|x_i - \tau(x_i)1 - \xi\|_2 \rightarrow 0.$$

Choose an L^2 -dense family

$$\{a_\alpha : \alpha < \kappa\} \subset M_{\text{sa}} \cap H_0(M).$$

By Lemma 9.2, choose a separable diffuse abelian von Neumann subalgebra $N_0 \subset M$ and an orthonormal basis $B_0 \subset \mathcal{S}_0(N_0)$ of $H_0(N_0)$.

We recursively construct von Neumann algebras $N_\alpha \subset M$ and orthonormal families $B_\alpha \subset \mathcal{S}_0(M)$, for $\alpha < \kappa$, such that:

- (i) $B_\alpha \subset \mathcal{S}_0(N_\alpha)$ is an orthonormal basis of $H_0(N_\alpha)$;
- (ii) $a_\alpha \in N_{\alpha+1}$;
- (iii) if $\alpha < \beta$, then $N_\alpha \subset N_\beta$ and $B_\alpha \subset B_\beta$;
- (iv) $\text{dens } L^2(N_\alpha) \leq \max\{\aleph_0, |\alpha|\}$.

The initial pair is (N_0, B_0) .

Suppose N_α, B_α have been constructed. Put

$$\lambda_\alpha = \max\{\aleph_0, |\alpha|\}.$$

Since $\alpha < \kappa$ and κ is identified with its initial ordinal, $\lambda_\alpha < \kappa$. Proposition 9.1, applied to $N = N_\alpha$ and $X = \{a_\alpha\}$, gives a von Neumann algebra $N_{\alpha+1}$ containing $N_\alpha \cup \{a_\alpha\}$ and an orthonormal family $U_\alpha \subset \mathcal{S}_0(N_{\alpha+1})$ such that

$$B_\alpha \cup U_\alpha$$

is an orthonormal basis of $H_0(N_{\alpha+1})$. Set

$$B_{\alpha+1} = B_\alpha \cup U_\alpha.$$

The density bound is also supplied by Proposition 9.1.

At a limit ordinal $\delta < \kappa$, set

$$N_\delta = W^*\left(\bigcup_{\alpha < \delta} N_\alpha\right), \quad B_\delta = \bigcup_{\alpha < \delta} B_\alpha.$$

In a finite von Neumann algebra,

$$L^2(N_\delta) = \overline{\bigcup_{\alpha < \delta} L^2(N_\alpha)}^{\|\cdot\|_2},$$

by the same Kaplansky-density argument as in Lemma 8.1. We claim that

$$H_0(N_\delta) = \overline{\bigcup_{\alpha < \delta} H_0(N_\alpha)}^{\|\cdot\|_2}.$$

Let $\xi \in H_0(N_\delta)$ and let $\varepsilon > 0$. By the displayed L^2 -continuity and by taking self-adjoint parts, there are $\alpha < \delta$ and $x = x^* \in N_\alpha$ such that

$$\|x - \xi\|_2 < \varepsilon.$$

Since $\xi \perp 1$ and $\|1\|_2 = 1$,

$$|\tau(x)| = |\langle x - \xi, 1 \rangle_2| \leq \|x - \xi\|_2 < \varepsilon.$$

Therefore $x - \tau(x)1 \in H_0(N_\alpha)$, and

$$\|x - \tau(x)1 - \xi\|_2 \leq \|x - \xi\|_2 + |\tau(x)| \|1\|_2 < 2\varepsilon.$$

This proves the claim. Since the bases are nested, B_δ is an orthonormal basis of $H_0(N_\delta)$. Also

$$\text{dens } L^2(N_\delta) \leq \max\{\aleph_0, |\delta|\} < \kappa.$$

Put

$$B = \bigcup_{\alpha < \kappa} B_\alpha.$$

Since the B_α 's are nested orthonormal families, B is orthonormal, and $B \subset \mathcal{S}_0(M)$. For every $\alpha < \kappa$, the element a_α belongs to $N_{\alpha+1}$ and has trace zero; hence

$$a_\alpha \in H_0(N_{\alpha+1}) \subset \overline{\text{span}_{\mathbb{R}} B}.$$

Since the family $(a_\alpha)_{\alpha < \kappa}$ is L^2 -dense in $H_0(M)$, the orthonormal family B is a Hilbert orthonormal basis of $H_0(M)$. Hence $|B| = \text{dens } H_0(M) = \kappa$, because the density character of an infinite Hilbert space equals the cardinality of any Hilbert orthonormal basis. By Lemma 3.2, $\{\widehat{1}\} \cup \{\widehat{s} : s \in B\}$ is a Hilbert orthonormal basis of $L^2(M, \tau)$ consisting of self-adjoint unitaries. Since $\kappa > \aleph_0$, this basis has cardinality κ . Each of these vectors is a trace vector, again by Lemma 3.2. $\square$

Remark 9.4. The nonseparable argument uses the strict density gap $\text{dens } L^2(N) < \text{dens } L^2(M)$ in Lemma 4.4. This is why the proof is organized as a transfinite chain of subalgebras of smaller density character. The separable case is handled in [HTZ26] by a finite-dimensional norming argument and is not used in the present proof.

10. CONSEQUENCE TOGETHER WITH THE SEPARABLE CASE

Corollary 10.1 (Kadison's trace-vector basis problem). *Let M be a type II_1 factor with faithful normal tracial state τ . Then $L^2(M, \tau)$ admits a Hilbert orthonormal basis whose vectors are represented by self-adjoint unitaries in M . Consequently $L^2(M, \tau)$ admits a Hilbert orthonormal basis consisting of trace vectors for the left action of M .*

Proof. If $\text{dens } L^2(M, \tau) = \aleph_0$, this is the separable case proved in [HTZ26]. If $\text{dens } L^2(M, \tau) > \aleph_0$, this is Theorem 9.3. In both cases the basis vectors are represented by self-adjoint unitaries in M . For every self-adjoint unitary $s \in M$ and every $x \in M$,

$$\langle x\widehat{s}, \widehat{s} \rangle_2 = \tau(ssx) = \tau(xss) = \tau(x),$$

so each basis vector is a trace vector. $\square$

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